

(0000CJA102120024)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Main)

AIOT

29-01-2021

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE**ANSWER KEY****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	C	B	B	A	A	C	D	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	A	C	D	A	B	A	C	D
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	225.00	14.60	2.00	5.96 to 5.98	9.00	15.85 to 15.87	10.00	2.25	200.00	6.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	A	A	A	C	B	A	D	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	B	C	D	A	D	C	D	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	13.00	33.33	8.00	7.00	64.59	0.88	2.70 TO 2.74	1.75	1.25	3.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	C	C	D	B	D	C	B	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	A	B	A	D	B	B	A	A	D
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	2.00	0.66 or 0.67	5.00	0.25	0.75	0.60	3.00	0.00	6.00	2.00



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PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (C)

Sol. (1) 2.3056 → 4 decimals
 10.138 → 3 decimals
 -7.4671 → 4 decimals

4.9765

Answer should have 3 decimals.

(2) $2.38 \times 1.0 = 2.38 \rightarrow$ answer should have 2 significant digits

(3) $\frac{8.05}{3.1} = 2.59 \approx 2.6$ answer should have 2SD

(4) $1.11 - 0.1 = 1.01 \approx 1.0$ but it is an intermediate step so we keep 1 digit extra.
 $1.01 \times 9.0 = 9.09 \approx 9.1 \rightarrow$ both 1.01 and 9.0 have 2SD

2. Ans. (B)

Sol. $\tan \theta = \frac{v_y}{v_x} = \frac{u_y - gt}{u_x}$

$$u_x = \frac{4}{1} \text{ m/s}$$

$$3 = u_y \times 1 - \frac{1}{2} \times 10 \times 1^2$$

$$u_y = 8 \text{ m/s}$$

$$\Rightarrow \tan \theta = \left| \frac{8-10}{4} \right| = \left| -\frac{1}{2} \right|$$

3. Ans. (C)

Sol. $a = \frac{v_0^2}{R}$

$$\text{after 10 kicks, } v = v_0 + \frac{10v_0}{100} = 1.1v_0$$

$$a = \frac{(1.1v_0)^2}{R_1} = \frac{v_0^2}{R}$$

$$\Rightarrow R_1 = 1.21 R$$

4. Ans. (B)

$$\begin{aligned} \text{Sol. } I &= \frac{m}{4} \times \frac{a^2}{3} \times 2 + \frac{m}{4} \times \left(\frac{a^2}{12} + \left(\frac{a}{2} \right)^2 + a^2 \right) \times 2 \\ &= \frac{ma^2}{6} + \frac{m}{2} \left(\frac{a^2 + 3a^2 + 12a^2}{12} \right) \\ &= \frac{ma^2}{6} + \frac{2}{3} ma^2 = \frac{ma^2 + 4ma^2}{6} = \frac{5ma^2}{6} \end{aligned}$$

5. Ans. (B)

$$\text{Sol. } T = 2\pi \sqrt{\frac{\ell}{g}}$$

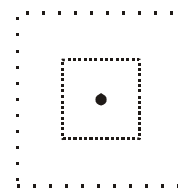
In summer, length will increase due to expansion, hence for same time period we have to decrease ℓ in summer.

6. Ans. (A)

$$\begin{aligned} \text{Sol. } \frac{dH}{dt} &= -2\pi x^2 k \frac{dT}{dx} \\ \Rightarrow \frac{dH}{dt} \times \int_{2.5}^3 \frac{dx}{x^2} &= 2\pi k \int_{-40}^5 dT \\ \frac{dH}{dt} &= \frac{2\pi \times k \times 45}{\frac{1}{2.5} - \frac{1}{3}} = 2.25\pi kW \end{aligned}$$

7. Ans. (A)

$$\begin{aligned} \text{Sol. } \lambda &= \frac{v}{\sqrt{2}\pi d^2 n N_a} = \frac{RT}{\sqrt{2}\pi d^2 p N_a} \\ &= \frac{\frac{25}{3} \times 300}{\sqrt{2} \times \pi \times 10^{-20} \times 10^5 \times 6 \times 10^{23}} \\ &= \frac{25}{6\sqrt{2}\pi} \times 10^{-6} \text{ m} \\ x^3 &= \frac{v}{n N_a} \\ \Rightarrow x &= \left(\frac{RT}{p N_a} \right)^{1/3} \\ &= \left(\frac{\frac{25}{3} \times 300}{10^5 \times 6 \times 10^{23}} \right)^{1/3} \end{aligned}$$



8. Ans. (C)

Sol. $v = 108 \times \frac{5}{18} = 30 \text{ m/s}$

$$f_1 = \frac{C}{C-v} f_0$$

$$550 = \frac{330}{330-30} f_0$$

$$f_0 = 500 \text{ Hz}$$

$$f = \frac{330}{330+30} \times 500 = \frac{33}{36} \times 500 = \frac{5500}{12}$$

9. Ans. (D)

Sol. $B = mg \Rightarrow B_1 = B_2$

$$\rho_1 V_1 g = B = \rho_w V_{LD} g$$

$$\frac{\rho_1}{\rho_w} = \frac{V_{LD}}{V_1} = f_1$$

$$\frac{\rho_2}{\rho_w} = f_2$$

$$\Rightarrow \rho_2 > \rho_1$$

10. Ans. (D)

Sol. $v = \frac{2 r^2 g (\rho - \sigma)}{9 \eta}$

$$v' = \frac{2}{9} r^2 g \frac{(\rho - 2\sigma)}{\eta}$$

11. Ans. (B)

Sol. Field lines are proportional to quantity of charge & they originate at positive charge & terminates at -ve charge or at ∞

12. Ans. (B)

Sol. $i = \frac{v}{R+G}$

$$\frac{i}{2} = \frac{\frac{v}{SG} + R}{\frac{S}{S+G} + R} \times \frac{S}{S+G}$$

$$\Rightarrow \frac{2S}{SG+RG+RS} = \frac{1}{R+G}$$

$$SG+SR=RG$$

$$\frac{SR}{R-S} = G$$

$$R \gg S$$

$$\Rightarrow G \approx S$$

13. Ans. (A)

Sol. $WD = \Delta QV$

$$\Rightarrow \Delta Q = 4\mu C$$

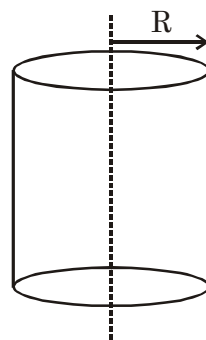
$$Q_i = \frac{2 \times 4}{2+4} \times 6 = 8\mu C$$

$$Q_f = 12\mu C = C_{eq} \times 6$$

$$C_{eq} = 2 \mu F \Rightarrow C_1 = C_2 = 4 \mu F$$

$$k = \frac{C_1}{C_0} = 2$$

14. Ans. (C)



Sol.

Applying Ampere's Law

For inside

$$B2\pi r = \mu_0 J \pi r^2$$

$$\Rightarrow B \propto r$$

For outside

$$B2\pi r = \mu_0 J \pi R^2$$

$$\Rightarrow B \propto \frac{1}{r}$$

15. Ans. (D)

Sol. Heat will develop due to eddy current which will oppose the motion and hence a force is required.

16. Ans. (A)

Sol. Low pass filter will allow low frequency signal to pass while high pass filter allow high frequency to pass through.

17. Ans. (B)

Sol. This experiment is the famous Davission-Germer experiment which establishes wave nature of electrons.

18. Ans. (A)

Sol. $Q_{(i)} = (M_{Br} - M_{Kr}) c^2 > 0$

$Q_{(ii)} = (M_{Br} - M_{Se} - 2M_e) c^2 > 0$

$Q_{(ii)} = (M_{Br} - M_{Se}) c^2 > 0$

19. Ans. (C)

Sol. $i_{\max} = \frac{P_{\max}}{V_z} = \frac{3}{100}$

$\therefore R_{s_{\min}} = \frac{V_{RS}}{i_{\max}}$

$= \frac{3}{3/100} = 100\Omega$

20. Ans. (D)

Sol. $\frac{\lambda_1}{w} = \frac{3 \times 540}{2w}$

$\lambda_1 = 810 \text{ nm}$

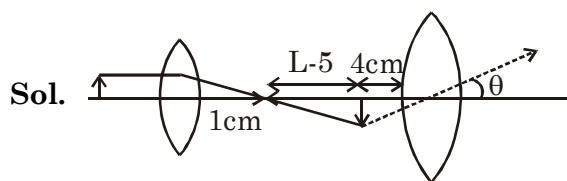
SECTION-II

1. Ans. 225.00

Sol. $v = \frac{\omega}{k} = \frac{5\pi \times 30}{\pi} = 150 = \sqrt{\frac{T}{\frac{1}{100}}}$

$\Rightarrow T = \frac{2250000}{100} = 225N$

2. Ans. 14.6



for better resolution, smaller focal length should be objective.

$h_0 = 1 \text{ cm } f_e = 4\text{cm.}$

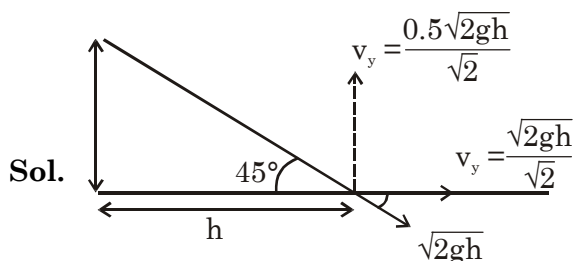
$Q_0 = \frac{h_0}{25}$

$Q = \frac{60h_0}{25} = \frac{h_1}{4}$

$\frac{L-5}{1} = \frac{h_1}{h_0} = \frac{240}{25}$

$L = 14.6 \text{ cm}$

3. Ans. 2.00

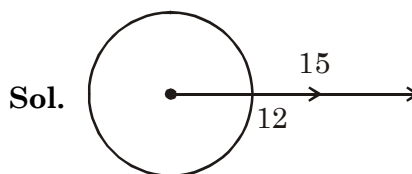


$R = u_x \times T$

$= \sqrt{gh} \times \frac{0.5\sqrt{2gh} \times 2}{g} = h$

$\Rightarrow AB = h + h = 2h$

4. Ans. 5.96 to 5.98



$\frac{kQe}{0.12} + \frac{1}{2}mu^2 = \frac{kQe}{0.15} + \frac{1}{2}mu^2$

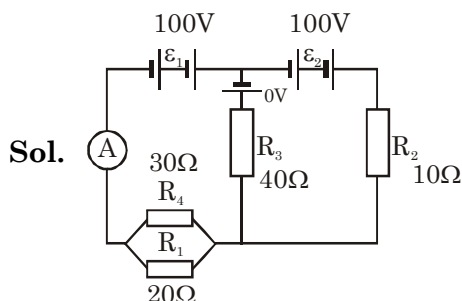
$\frac{kQe \times 0.03}{0.12 \times 0.15} = \frac{1}{2}m \times (v^2 - u^2)$

$\frac{9 \times 10^9 \times Q \times 1.6 \times 10^{-19}}{0.12 \times 0.15} \times 0.03 = \frac{1}{2} \times 4 \times 10^{10} \times 99$

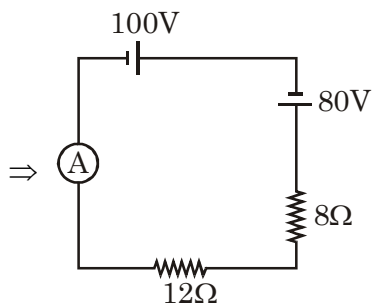
$Q = \frac{9.1 \times 99 \times 10^{-11}}{12}$

$= \frac{75.075 \times 10^{-12}}{4\pi \times (0.1)^2} = 5.97 \text{ nC/m}^2$

5. Ans. 9.00



$\epsilon_{eq} = \frac{\frac{0}{40} + \frac{100}{10}}{\frac{1}{40} + \frac{1}{10}} = \frac{10}{\frac{5}{40}} = 80V$

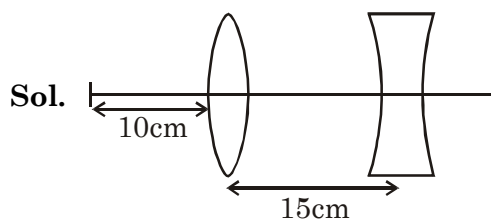


$$R_{eq} = \frac{400}{50} = 8\Omega$$

$$R'_{eq} = \frac{30 \times 20}{30 + 20} = \frac{600}{50} = 12$$

$$i = \frac{180}{12 + 8} = 9A$$

6. Ans. 15.85 to 15.87



$$\frac{1}{u} + \frac{1}{-10} = \frac{1}{35}$$

$$\frac{1}{u} = \frac{1}{35} - \frac{1}{10} = \frac{2-7}{70} = \frac{-1}{14}$$

$$\frac{1}{u} + \frac{1}{+29} = \frac{-1}{35}$$

$$\frac{1}{u} = \frac{-1}{35} - \frac{1}{29} = \frac{-64}{35 \times 29}$$

$$u = \frac{35 \times 29}{64} = 15.85 \text{ to } 15.86$$

7. Ans. 10.00

$$\text{Sol. } 400 \times 1 \times 12.5 = 500 \times 5 \times (100 - 36.5)$$

$$S = \frac{10}{63.5} \text{ cal/gm } ^\circ\text{C}$$

$$1\text{gm} = \frac{1}{63.5} \text{ mol}$$

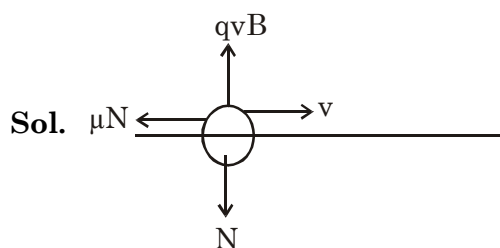
$$S = 10 \text{ cal/mol } ^\circ\text{C}$$

8. Ans. 2.25

$$\text{Sol. } 0^2 = v^2 - 2\mu_k g S_{rel}$$

$$\frac{9}{2 \times 2} = S_{rel} = 2.25\text{m}$$

9. Ans. 200.00



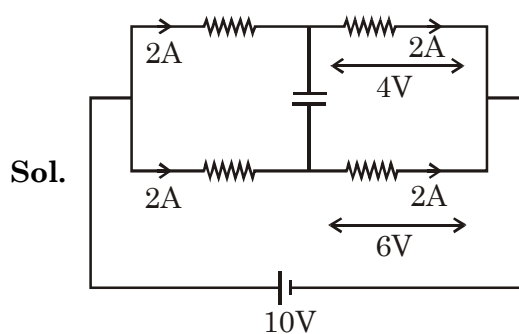
$$N = qvB$$

$$-\mu qvB = \frac{mdv}{dt} = mv \frac{dv}{dS}$$

$$\mu qBS = mv$$

$$S = \frac{3 \times 10^{-6} \times 4}{0.3 \times 10^{-6} \times 0.2} = 200\text{m}$$

10. Ans. 6.00



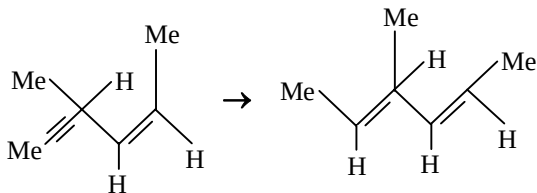
Sol.

$$P_d = 6 - 4 = 2 \text{ V} \Rightarrow Q = CV = 6\mu\text{C}$$

SECTION-I

1. **Ans.(B)**

Addition on triple bond takes place by the addition of hydrogen



Will have a plane of symmetry and hence optically inactive. So option (2) is correct.

2. **Ans.(C)**

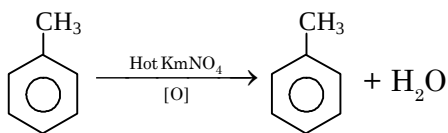
3. **Ans.(A)**

$$G \frac{\ell}{a} = \kappa \quad \dots (1)$$

$$\Lambda_m = \kappa \times \frac{1000}{n / (\ell \times a) \times 10^{-3}} \quad \dots (2)$$

from (1) & (2) $\Lambda_m = 2\Omega^{-1} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$

4. **Ans.(A)**



Benzoic acid

5. **Ans.(A)**

6. **Ans.(C)**

$$i = \frac{m_{\text{theor}}}{m_{\text{app}}} = \frac{30}{m_{\text{app}}}$$

$$\Delta T_b = i k_b \cdot m \quad \text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$$

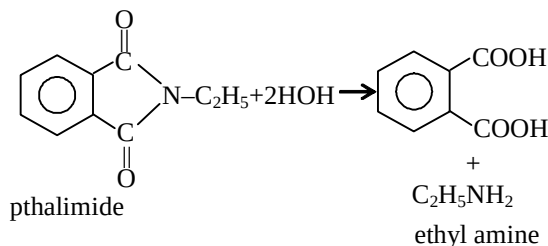
$$0.0156 = \left(\frac{m \times x}{m} \right) \times 0.52 \times m$$

$$m + x = 0.03$$

$$10^{-2} = \frac{x^2}{m - x} = \frac{x^2}{0.03 - x - x} = \frac{x^2}{0.03 - 2x}$$

$$\Rightarrow x = 0.01 \Rightarrow i = 1.5 = \frac{30}{m_{\text{app}}} \Rightarrow m_{\text{app}} = 20$$

7. **Ans.(B)**



Therefore option (B) is correct.

8. **Ans.(A)**

9. **Ans.(D)**

$$Z = \frac{PV_m}{RT} \Rightarrow 0.5 = \frac{24 \times V_m}{24}$$

$$V_m = 0.5 \text{ l/mole}$$

$$\left(P + \frac{a}{V_m^2} \right) (V_m - b) = RT$$

$$\left(24 + \frac{a}{0.5^2} \right) (0.5 - 0.1) = 24$$

$$24 + \frac{a}{0.5^2} = 60$$

$$\frac{a}{0.5^2} = 36$$

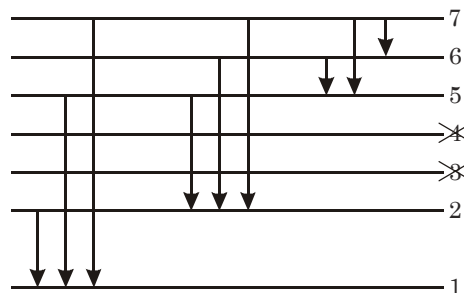
$$a = 36 \times 0.5^2 = 9 \frac{\text{atm} \cdot \ell^2}{\text{mol}^2}$$

10. **Ans.(C)**

In H_2O (polar solvent) dibromophenol derivative and in CS_2 (non-polar) solvent momeobromo phenol derivative is obtained.

11. **Ans.(A)**

12. **Ans.(C)**



13. Ans.(B)

Sucrose D(+) glucose + D(-) Fructose

$$[\alpha] = +66.5^\circ \quad [\alpha] = +52.5^\circ \quad [\alpha] = -92.8^\circ$$

Thus rotation changes from positive to negative after hydrolysis. Due to this reason hydrolysis of sucrose is known as inversion and mixture after hydrolysis is known as invert sugar.

Therefore option (B) is correct.

14. Ans.(C)

15. Ans.(D)

$$\begin{array}{ccc} A(g) & \rightleftharpoons & 2B(g). \\ E_m & \frac{2\text{mol}}{V\ell} & \frac{4\text{mol}}{V\ell} \\ 2\text{mol} & \frac{2+2}{V} & \frac{4}{V} \\ A_{\text{added}} & & \\ \text{New } E_m & \frac{4-x}{V} & \frac{4+2x}{V} \\ V \times 2 & \frac{4-x}{2V} & \frac{4-2x}{2V} \\ \text{New } E_m & \frac{4-x-y}{2V} & \frac{4+2x+2y}{2V} \\ & = \frac{4-Z}{2V} & = \frac{4+2Z}{2V} \quad (Z=x+y) \end{array}$$

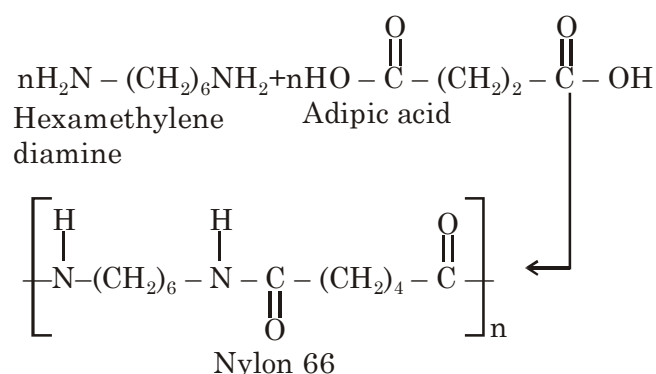
$$\text{Now, } K_c = \frac{[B]^2}{[A]} = \text{constant}$$

$$\text{or, } \left(\frac{4}{V}\right)^2 = \left(\frac{4+2Z}{2V}\right)^2 \Rightarrow Z = 1.29$$

$$\therefore \text{mole of B at new } E_m = 4 + 2Z = 6.58$$

16. Ans.(A)

Sol.



17. Ans.(D)

18. Ans. (C)

19. Ans.(D)

$$q = 0, w = 0, \Delta U = 0 = f(V, T)$$

As volume changes, temperature must change to maintain constant internal energy.

20. Ans. (C)

SECTION-II

1. Ans.(13.00)

2. Ans.(33.33)

$$t = \frac{t_{1/2}}{\log 2} \cdot \log \frac{V_\infty}{V_\infty - V_t}$$

$$20 = \frac{10}{\log 2} \cdot \log \frac{V_\infty}{V_\infty - 25}$$

$$\therefore V_\infty = 33.33 \text{ ml}$$

3. Ans. (8.00)

4. Ans.(7.00)

5. Ans.(64.59)

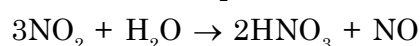
$$\sqrt{2}a = \frac{258.36}{\sqrt{2}}$$

$$\alpha = 258.36 = 2r$$

6. Ans.(0.88)

$$\text{Moles of NO} = \frac{600 \times 10^{-3} \times 100}{30} = 2$$

$$\therefore \text{moles of NO}_2 = 2$$



$$0.2\text{M} \quad \frac{0.4}{3}\text{M} \quad \text{or} \quad \text{pH} = -\log \frac{0.4}{3} = 0.88$$

7. Ans.(2.70 to 2.74)

Eq. = no. of faraday

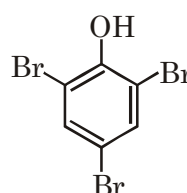
$$\frac{10 \times 0.2 \times 10^{-2} \times 19.7}{197} \times 3 = \frac{0.193 \times t}{96500}$$

8. Ans.(1.75)

9. Ans.(1.25)

10. Ans.(3.00)

Sol. P is



SECTION-I

1. Ans. (B)

$$\text{Sol. } T_{r+1} = {}^{15}C_r (x^2)^{15-r} \left(\frac{3\lambda}{x}\right)^r$$

$$T_{r+1} = {}^{15}C_4 (3\lambda)^r \cdot x^{30-3r}$$

$$30 - 3r = 21 \Rightarrow r = 3$$

$$T_4 = {}^{15}C_3 (3\lambda)^3$$

$${}^{15}C_3 \times 27 \times \lambda^3 = 3640 \Rightarrow \lambda^3 = \frac{3640}{27 \times 455}$$

$$\lambda^3 = \frac{8}{27} \Rightarrow \lambda = \frac{2}{3}$$

2. Ans. (D)

$$\text{Sol. } \int_0^{\pi/2} \left(\frac{1}{(1+\tan^2 x)(1+2^x)} + \frac{1}{(1+\tan^2 x)(1+2^{-x})} \right) dx$$

$$\int_0^{\pi/2} \frac{dx}{(1+\tan^2 x)} = \int_0^{\pi/2} \cos^2 x dx = \frac{\pi}{4}$$

3. Ans. (C)

$$\text{Sol. } 3 - 2^x - 2^{1-x} \geq 0 \quad \text{put } 2^x = t$$

$$3 - t - \frac{2}{t} \geq 0 \Rightarrow t^2 - 3t + 2 \leq 0$$

$$(t-1)(t-2) \leq 0 \Rightarrow t \in [1, 2]$$

$$x \in [0, 1]$$

4. Ans. (C)

$$\text{Sol. } f(x) = x^2 + x + \left(\frac{\lambda+1}{\lambda}\right)$$

$$f(1) < 0 \Rightarrow 2 + \left(\frac{\lambda+1}{\lambda}\right) < 0$$

$$\frac{3\lambda+1}{\lambda} < 0 \Rightarrow \lambda \in \left(-\frac{1}{3}, 0\right)$$

5. Ans. (D)

$$\text{Sol. } P = 1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \frac{14}{3^4} + \dots + \infty$$

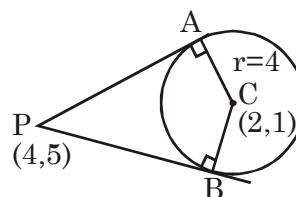
$$\frac{P}{3} = \frac{1}{3} + \frac{2}{3^2} + \frac{6}{3^3} + \frac{10}{3^4} + \dots + \infty$$

$$\frac{2P}{3} = 1 + \frac{1}{3} + \left[\frac{4}{3^2} + \frac{4}{3^3} + \frac{4}{3^4} + \dots + \infty \right]$$

$$\frac{2P}{3} = \frac{4}{1 - \frac{1}{3}} \Rightarrow P = 3$$

6. Ans. (B)

Sol.



$$L = \sqrt{16 + 25 - 16 - 10 - 11} = 2$$

$$\text{area of quad PACB} = LR = 8$$

7. Ans. (D)

$$\text{Sol. } f(x) = \begin{cases} -\frac{1}{x} & x \in (-\infty, -1] \\ ax^2 + b & x \in (-1, 1) \\ \frac{1}{x} & x \in [1, \infty) \end{cases}$$

$$\text{cont at } x = 1, a + b = 1$$

$$\text{diff at } x = 1, 2a = -1 \Rightarrow a = -\frac{1}{2} \Rightarrow b = \frac{3}{2}$$

8. Ans. (C)

$$\text{Sol. } \sim(p \rightarrow q) = p \wedge \sim q$$

A quadrilateral is a square and it is not a rhombus

9. Ans. (B)

Sol. Variance of 2001, 2003, 2006, 2007, 2009, 2010 is same as variance of 1, 3, 6, 7, 9, 10

$$\text{mean} = \frac{1+3+6+7+9+10}{6} = 6$$

$$\text{var}(x) = \frac{1^2+3^2+6^2+7^2+9^2+10^2}{6} - 36$$

$$\text{var}(x) = 10$$

10. Ans. (D)

$$\text{Sol. } [\vec{a} \ \vec{b} \ \vec{c}] = 0$$

$$\begin{vmatrix} 1 & 6 & 3 \\ 3 & 2 & 1 \\ \alpha+1 & \beta-1 & 1 \end{vmatrix} = 0$$

$$8\beta - 24 = 0 \Rightarrow \beta = 3$$

$$|\vec{c}| = \sqrt{6}$$

$$(\alpha+1)^2 + (\beta-1)^2 + 1 = 6 \Rightarrow (\alpha+1)^2 = 1$$

$$\alpha = 1 = 1, -1 \Rightarrow \alpha = -2, 0$$

$$\alpha + \beta = 1, 3$$

11. Ans. (A)

Sol. $y^2 = x$, $a = \frac{1}{4}$

$$x^2 = -y, b = \frac{1}{4}$$

$$A = \frac{16|ab|}{3} = \frac{1}{3}$$

12. Ans. (A)

Sol. $x dy + y dx = \frac{xy dx}{\sqrt{1-x^2}}$

$$\frac{d(xy)}{xy} = \frac{dx}{\sqrt{1-x^2}}$$

Integrate both sides we get
 $\log_e xy = \sin^{-1}x + C$

13. Ans. (B)

Sol. $\frac{x}{4} = \cos t - \sin t \quad \dots (1)$

$$\frac{y}{5} = \cos t + \sin t \quad \dots (2)$$

$(1)^2 + (2)^2$ gives

$$\left(\frac{x}{4}\right)^2 + \left(\frac{y}{5}\right)^2 = 2$$

14. Ans. (A)

Sol. If the two lines are perpendicular then $a = 2$

$$D = 0 \Rightarrow c = -3$$

$$(D = abc + 2fgh - af^2 - bg^2 - ch^2)$$

15. Ans. (D)

Sol. EHINTZ

words starting with E is 5!

words starting with H is 5!

words starting with I is 5!

words starting with N is 5!

words starting with T is 5!

Z	E	H			
---	---	---	--	--	--

 3!

Z	E	I			
---	---	---	--	--	--

 3!

Z	E	N	H		
---	---	---	---	--	--

 2!

Z	E	N	I	H	
---	---	---	---	---	--

 1!

Z	E	N	I	T	H
---	---	---	---	---	---

 1!

16. Ans. (B)

$A = \{1, 2, 3, 4\}$. As $(4, 4) \notin R$ so not reflexive
 Also not transitive

17. Ans. (B)

Sol. $|A - xI| = \begin{vmatrix} 1/2 - x & 1/2 \\ a & b - x \end{vmatrix} = x^2 - \left(b + \frac{1}{2}\right)x + \frac{b-a}{2}$

from Cayley-Hamilton Theorem

$$A^2 - \left(b + \frac{1}{2}\right)A + \left(\frac{b-a}{2}\right)I = 0$$

$$\Rightarrow A^3 = \left(b + \frac{1}{2}\right)A^2 - \left(\frac{b-a}{2}\right)A$$

$$\Rightarrow A^3 = \left(b + \frac{1}{2}\right)\left\{\left(b + \frac{1}{2}\right)A - \left(\frac{b-a}{2}\right)I\right\} - \left(\frac{b-a}{2}\right)A$$

$$\Rightarrow A^3 = \left\{\left(b + \frac{1}{2}\right)^2 - \left(\frac{b-a}{2}\right)\right\}A - \left(b + \frac{1}{2}\right)\left(\frac{b-a}{2}\right)I$$

$$\therefore \left(b + \frac{1}{2}\right)^2 - \left(\frac{b-a}{2}\right) = 1 \text{ \& } \left(b + \frac{1}{2}\right)\left(\frac{b-a}{2}\right) = 0$$

$$\therefore (a, b) = \left(\frac{1}{2}, \frac{1}{2}\right), \left(-\frac{3}{2}, -\frac{3}{2}\right), \left(\frac{3}{2}, -\frac{1}{2}\right)$$

18. Ans. (A)

Sol. Line L is $\frac{x-1}{1} = \frac{y-2}{2\sqrt{2}} = \frac{z}{0}$

This line L makes an angle of 45° with the plane $\sqrt{2}x + y - z = 1$

$\therefore$ Required distance PQ is perpendicular distance of plane from P is i.e.,

$$PQ = \frac{|\sqrt{2} + 7 - \sqrt{2} - 1|}{\sqrt{2+1+1}} = 3$$

19. Ans. (A)

Sol. $\int e^{\sin x} \left(\frac{5 + \cos^2 x}{(2 - \sin x)\sqrt{3 + \cos^2 x}} \right) \cos x dx$

put $\sin x = t$

$$\int e^t \left(\frac{6 - t^2}{(2 - t)\sqrt{4 - t^2}} \right) dt$$

$$\int e^t \left(\frac{4 - t^2}{(2 - t)\sqrt{4 - t^2}} + \frac{2}{(2 - t)\sqrt{4 - t^2}} \right) dt$$

$$\int e^t \left(\sqrt{\frac{2+t}{2-t}} + \frac{2}{(2-t)^{3/2}(2+t)^{1/2}} \right) dt$$

$$\text{If } g(t) = \sqrt{\frac{2+t}{2-t}}, g'(t) = \frac{2}{(2-t)^{3/2}(2+t^2)^{1/2}}$$

$$e^t \sqrt{\frac{2+t}{2-t}} + c$$

$$f(x) = e^{\sin x} \sqrt{\frac{2+\sin x}{2-\sin x}} + c$$

$$f(0) = 1 \Rightarrow c = 0$$

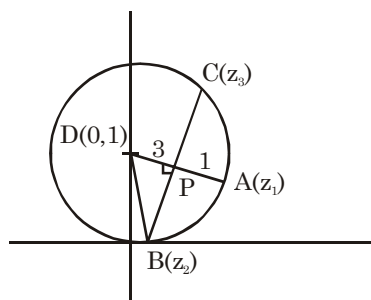
$$f(x) = e^{\sin x} \sqrt{\frac{2+\sin x}{2-\sin x}}$$

$$f\left(\frac{\pi}{2}\right) = \sqrt{3}e$$

20. Ans. (D)

Sol. $\frac{3z_1 + i}{4} = \frac{2z_2 + 2z_3}{4}$

P = point of intersection of AD & BC



$$AD = 1 \Rightarrow DP = \frac{3}{4}$$

$$BP = \sqrt{1 - \frac{9}{16}} = \frac{\sqrt{7}}{4} \Rightarrow BC = \frac{\sqrt{7}}{2}$$

$$\begin{aligned} \text{area of Quad. ABCD} &= \frac{1}{2} \cdot AD \times BC \\ &= \frac{1}{2} \times 1 \times \frac{\sqrt{7}}{2} = \frac{\sqrt{7}}{4} \end{aligned}$$

SECTION-II

1. Ans. 2.00

$$\frac{dy}{dx} = 3ax^2 - 2bx$$

$$\left. \frac{dy}{dx} \right|_{x=1} = 3a - 2b = 3$$

$$a = \frac{2b+3}{3} \in [1, 2]$$

$$2b+3 \in [3, 6]$$

$$2b \in [0, 3]$$

$$b \in \left[0, \frac{3}{2}\right]$$

2. Ans. 0.66 or 0.67

$$\frac{P(A \cap B)}{P(B)} = \frac{1}{4}, \frac{P(A \cap B)}{P(A)} = \frac{1}{12}$$

$$\text{Divide both } \frac{P(B)}{P(A)} = \frac{1}{3}$$

$$P(A) - P(A \cap B) = \frac{1}{12} \Rightarrow P(A) - \frac{P(A)}{12} = \frac{1}{12}$$

$$P(A) = \frac{1}{11}, P(B) = \frac{1}{33}$$

3. Ans. 5.00

$$\begin{vmatrix} 1-\alpha & 6 & 6 \\ 4 & -1-\alpha & 4 \\ 2\alpha & 2\alpha & \alpha-5 \end{vmatrix} = 0$$

$$\alpha = -5$$

4. Ans. 0.25

$$g(f(x)) = x \Rightarrow g'(f(x)) \cdot f'(x) = 1$$

$$f(x) = 1 \Rightarrow x = 0$$

$$g'(1) \cdot f'(0) = 1$$

$$f'(x) = 2x + e^{-x}$$

$$f'(0) = 1$$

$$g'(1) = 1$$

5. Ans. 0.75

Sol. $x_1 = \lim_{x \rightarrow \infty} \frac{2^{\frac{x^n}{e^x}} - 3^{\frac{x^n}{e^x}}}{\frac{x^n}{e^x}}, \text{ put } \frac{x^n}{e^x} = t$

$$x_1 = \lim_{t \rightarrow 0} \frac{2^t - 3^t}{t} = \log_e \frac{2}{3}$$

$$x_2 = \lim_{x \rightarrow \infty} \frac{\cot^{-1}(\sqrt{x+1} - \sqrt{x})}{\sec^{-1}\left(\left(\frac{2x+1}{x-1}\right)^x\right)}$$

$$x_2 = \lim_{x \rightarrow \infty} \frac{\tan^{-1}\left(\frac{1}{\sqrt{x+1} - \sqrt{x}}\right)}{\sec^{-1}(\infty)}$$

$$x_2 = \frac{2}{\pi} \lim_{x \rightarrow \infty} \tan^{-1}(\sqrt{x+1} + \sqrt{x})$$

$$x_2 = \frac{2}{\pi} \cdot \frac{\pi}{2} = 1$$

6. Ans. 0.60

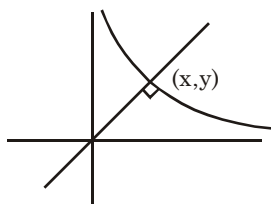
Sol. $OP^2 = x^2 + y^2$

$y = e^x, y' = e^x,$

slope of normal = $-\frac{1}{e^x}$

$\frac{y}{x} = -\frac{1}{e^x}$

$-\frac{1}{e^x} = \frac{e^x}{x} \Rightarrow -x = e^{2x}$



By hit and trial we get $x = -\frac{2}{5}$

$P \equiv \left(-\frac{2}{5}, e^{-2/5}\right)$

$OP = \sqrt{\frac{4}{25} + e^{-4/5}} \Rightarrow OP^2 = \frac{14}{25} = \frac{m}{n}$

7. Ans. 3.00

Sol. $f(x) = f(6-x) \Rightarrow f'(x) = -f'(6-x) \dots (1)$

put $x = 0, 2, 5$

$f'(0) = f'(6) = f'(2) = f'(4) = f'(5) = f'(1) = 0$

and from equation (1) we get $f'(3) = -f'(3)$

$\Rightarrow f'(3) = 0$

So $f'(x) = 0$ has minimum 7 roots in

$x \in [0, 6] \Rightarrow f''(x)$ has min 6 roots in $x \in [0, 6]$

$h(x) = f'(x) \cdot f''(x)$

$h'(x) = (f''(x))^2 + f'(x) f'''(x)$

$h(x) = 0$ has 13 roots in $x \in [0, 6]$

$h'(x) = 0$ has 12 roots in $x \in [0, 6]$

8. Ans. 0.00

$x^2(y+z)y^2(z+x)z^2(x+y) = a^3b^3c^3 = x^3y^3z^3$

$\Rightarrow (x+y)(y+z)(z+x) = xyz$

$\Rightarrow x^2(y+z) + y^2(z+x) + z^2(x+y) + xyz = 0$

$\Rightarrow a^3 + b^3 + c^3 + abc = 0$

9. Ans. 6.00

Sol. $x = t^2 - t + 1 \dots (1)$

$y = t^2 + t + 1 \dots (2)$

$y - x = 2t$ & $x + y = 2(t^2 + 1)$

_____ on eliminating 't' we get

$\Rightarrow (x+y-2) = 2\left(\frac{y-x}{2}\right)^2$

$(x-y)^2 = 2(x+y-2)$

Axis : $x - y = 0$

Tangent at vertex : $x + y - 2 = 0$

Vertex : $(1, 1) = (x, y)$

10. Ans. 2.00

Sol. $f(A, B) = \frac{1 - \sin B + \sin A - \sin A \sin B + \cos A \cos B}{\cos A(1 - \sin B)}$

$\Rightarrow f(A, B) = \frac{\cos A}{1 - \sin A} + \frac{1 + \sin B}{\cos B}$

$f(A, B) = \frac{\cos A \cos B + 1 + \sin B - \sin A - \sin A \sin B}{(1 - \sin A) \cos B}$

$\Rightarrow f(A, B) (\cos A - \cos A \sin B - \cos B + \sin A \cos B)$
 $= 2 \sin A - 2 \sin B$

$\Rightarrow f(A, B) = \frac{2(\sin A - \sin B)}{\cos A - \cos B + \sin(A - B)} = 2g(A, B)$